Janadri Yalla Yashwanth

Computer Vision / Machine Learning Engineer

yallayashwanth99@gmail.com | +91 8790819924 | LinkedIn | GitHub | Hyderabad, India 500001

SUMMARY

Computer-vision and ML engineer with hands-on production experience building deep-learning detection pipelines on drone and aerial imagery — SegFormer, SAM-2, DeepForest, and GroundingDINO/LangSAM — paired with geospatial post-processing (rasterio/GDAL, DSM/DTM elevation analysis). B.Tech in CSE (AI & ML), CGPA 8.6. Strong in PyTorch, OpenCV, and shipping models behind real APIs.

TECHNICAL SKILLS

Languages: Python, C, SQL
ML / Deep Learning: PyTorch, Hugging Face Transformers, TensorFlow, scikit-learn, CUDA/GPU
Computer Vision: SegFormer, SAM / SAM-2, DeepForest, GroundingDINO / LangSAM, CLIP, OpenCV, NumPy
Geospatial: rasterio, GDAL, GeoTIFF / COG, DSM/DTM & CHM, GeoJSON, CRS / projections
Backend / MLOps: FastAPI, Celery, Redis, Docker, GitHub Actions, REST APIs
Concepts: Semantic segmentation, object detection, NMS, model benchmarking, data preprocessing
Tools: Git & GitHub, Claude Code, VS Code, Jupyter, Hugging Face Hub, Docker, GitHub Actions, QGIS, Linux

EXPERIENCE

Software Engineer Trainee — Marut Drones *Feb 2026 – Present*

- Software engineer on the drone AI/automation team — shipped production computer-vision detection services, geospatial backend features, and autonomous-flight tooling; selected projects below.

Machine Learning Intern — Rooman Technologies *Jan 2025 – Mar 2025*

- Built ML solutions for healthcare; developed and evaluated a sleep-disorder classification model on physiological and behavioral data, achieving 98% accuracy.

KEY PROJECTS — MARUT DRONES

Aerial Image Object Detection — Multi-Model Benchmark

Python, PyTorch, Hugging Face Transformers, SegFormer, OpenCV

- Developed a computer-vision pipeline detecting buildings, vehicles, and trees from high-resolution drone imagery (stitched panoramas up to 14,400 x 7,200 px) using SegFormer-B5 (ADE20K, 150 classes) with PyTorch and Hugging Face Transformers.
- Benchmarked 4+ segmentation models (SegFormer-B5/B2, Aerial-Drone SegFormer, SAM + CLIP) head-to-head and selected ADE20K for best precision, cutting false positives ~3x versus over-segmenting alternatives.
- Engineered the full inference pipeline in OpenCV/NumPy: sliding-window tiling, HSV sky-skipping, semantic-mask-to-instance contour extraction, geometric shape filtering, and per-class Non-Max Suppression.
- Delivered annotated outputs (per-class count overlays + JSON), batch-folder processing, and drone-video support (frame-skip to annotated MP4), running on GPU/CUDA at ~7–11 s per image.

GeoAI Detection — Buildings & Trees from Drone Orthomosaics

Python, PyTorch, LangSAM (SAM + GroundingDINO), DeepForest, rasterio, OpenCV

- Built a geospatial building-detection pipeline on drone orthomosaics using zero-shot LangSAM (SAM + GroundingDINO) with a "building roof" text prompt, processing large GeoTIFFs block-wise via rasterio (no training data required).
- Eliminated flat false positives (roads, bare fields) by filtering every detection against the DSM — computing height = roof - ground (nDSM) and dropping objects without real vertical structure; exported georeferenced GeoJSON with area and height.
- Built individual-tree-crown detection with DeepForest (RetinaNet, NEON-pretrained), GSD-aware resampling tuned for dense tropical canopies, HSV green-filtering, crown merging, and NMS.
- Separated real trees from grass using a Canopy Height Model (CHM = DSM - DTM), keeping only crowns >= 1.5 m tall — removing flat green vegetation that color filters alone cannot reject.

Marut Survey Platform — Backend & AI Detection Services

FastAPI, Celery, Redis, PostgreSQL, PyTorch, SAM-2, S3/GDAL, Docker

- Built three production detection pipelines (stockpile, tree-crown, building) integrating three model families — SAM-2, DeepForest, and GroundingDINO/LangSAM — behind a FastAPI backend with async Celery/Redis job APIs (live progress, cancellation, owner-scoped RBAC).

- Derived per-object stockpile volumes (m3) and material classification from nDSM height integration, backed by a 7-factor confidence rubric and Union-Find cross-tile NMS dedup at 0.3 IoU.
- Streamed cloud-optimized GeoTIFFs block-by-block from S3 via GDAL /vsis3/ + rasterio (no full-raster load) and tuned S3 multipart transfers — 64 MB chunks, 25 concurrent threads, pooled connections — for multi-gigabyte orthomosaics.
- Packaged the PyTorch / SAM-2 / DeepForest / GroundingDINO stack into a Docker image shipped through GitHub Actions to systemd-managed GPU and CPU workers; Pydantic v2 schemas and Alembic migrations.

Flight Log Analyzer — Full-Stack Drone Diagnostics Platform

React, TypeScript, FastAPI, Python, Docker

- Built a full-stack web app that ingests raw drone flight logs and auto-generates professional PDF health/crash reports, supporting both ArduPilot (.bin) and JIYI K++ formats with automatic binary-vs-text detection via magic-byte sniffing.
- Engineered a from-scratch binary parser for the ArduPilot/DataFlash self-describing log format using Python struct, decoding 50+ message types and reconstructing real-world IST timestamps from GPS-week data.
- Implemented a phased diagnostic engine running 50+ automated checks (power, IMU/vibration, GPS, EKF, compass, RC, failsafes), classifying findings as PASS/WARN/FAIL and inferring probable crash cause.
- Developed automated PDF reporting with ReportLab + Matplotlib (flight-path maps, sensor charts, color-coded tables); built the React 19 + Vite + Tailwind front end with file upload and download handling.
- Containerized the full stack with Docker / Docker Compose; integrated a .NET converter via Mono + Xvfb inside Linux (~7,200 LOC, solo-built).

Autonomous Precision Landing on Moving Platform

ArduPilot, MAVLink, Raspberry Pi 5, AprilTag, OpenCV, Python

- Built an autonomous drone precision-landing system that lands on an AprilTag target on stationary and moving (ground-rover) platforms; achieved proven stationary-tag landings at ~5–10 cm accuracy.
- Wrote the Raspberry Pi 5 companion-computer controller (pupil-apriltags + picamera2 + OpenCV) streaming LANDING_TARGET and velocity setpoints to a CubeOrange (ArduCopter) over MAVLink (pymavlink) at 20 Hz.
- Designed a three-mode state machine — visual P-control with rover-velocity feed-forward, GPS-NAV haversine approach, and hover fallback — with pilot RC override, velocity caps, and touchdown auto-disarm.
- Implemented multi-vehicle MAVLink (drone + rover SYSIDs), camera calibration, systemd auto-start, and a BENCH safety mode for props-off bench testing.

EDUCATION

Sri Venkateswara College of Engineering, Tirupati *Dec 2021 – May 2025*

B.Tech in Computer Science (Artificial Intelligence & Machine Learning) — CGPA: 8.6

Narayana Junior College, Hyderabad *Jun 2019 – Mar 2021*

Intermediate (MPC) — 96%

CERTIFICATIONS

- CS50's Introduction to Programming with Python — Harvard University Online
- Databases: Topics in SQL — Stanford Online